

OpenStax Calculus Volume 2

Section 4.1 Exercises: Basics of Differential Equations

Source: <https://openstax.org/books/calculus-volume-2/pages/4-1-basics-of-differential-equations>.
Exercises reproduced from OpenStax, licensed under CC BY 4.0.

This document was converted programmatically from OpenStax content; formulas were translated automatically from MathML and may contain conversion errors — verify against the source before relying on them.

Determine the order of the following differential equations.

1. $y' + y = 3y^2$
2. $(y')^2 = y' + 2y$
3. $y''' + y''y' = 3x^2$
4. $y' = y'' + 3t^2$
5. $\frac{dy}{dt} = t$
6. $\frac{dy}{dx} + \frac{d^2y}{dx^2} = 3x^4$
7. $(\frac{dy}{dt})^2 + 8\frac{dy}{dt} + 3y = 4t$

Verify that the following functions are solutions to the given differential equation.

8. $y = \frac{x^3}{3}$ solves $y' = x^2$
9. $y = 2e^{-x} + x - 1$ solves $y' = x - y$
10. $y = e^{3x} - \frac{e^x}{2}$ solves $y' = 3y + e^x$
11. $y = \frac{1}{1-x}$ solves $y' = y^2$
12. $y = e^{x^2/2}$ solves $y' = xy$
13. $y = 4 + \ln x$ solves $xy' = 1$
14. $y = 3 - x + x \ln x$ solves $y' = \ln x$
15. $y = 2e^x - x - 1$ solves $y' = y + x$
16. $y = e^x + \frac{\sin x}{2} - \frac{\cos x}{2}$ solves $y' = \cos x + y$
17. $y = \pi e^{-\cos x}$ solves $y' = y \sin x$

Verify the following general solutions and find the particular solution.

18. Find the particular solution to the differential equation $y' = 4x^2$ that passes through $(-3, -30)$, given that $y = C + \frac{4x^3}{3}$ is a general solution.

19. Find the particular solution to the differential equation $y' = 3x^3$ that passes through $(1, 4.75)$, given that $y = C + \frac{3x^4}{4}$ is a general solution.
20. Find the particular solution to the differential equation $y' = 3x^2y$ that passes through $(0, 12)$, given that $y = Ce^{x^3}$ is a general solution.
21. Find the particular solution to the differential equation $y' = 2xy$ that passes through $(0, \frac{1}{2})$, given that $y = Ce^{x^2}$ is a general solution.
22. Find the particular solution to the differential equation $y' = (2xy)^2$ that passes through $(1, -\frac{1}{2})$, given that $y = -\frac{3}{C+4x^3}$ is a general solution.
23. Find the particular solution to the differential equation $y'x^2 = y$ that passes through $(1, \frac{2}{e})$, given that $y = Ce^{-1/x}$ is a general solution.
24. Find the particular solution to the differential equation $8\frac{dx}{dt} = -2\cos(2t) - \cos(4t)$ that passes through (π, π) , given that $x = C - \frac{1}{8}\sin(2t) - \frac{1}{32}\sin(4t)$ is a general solution.
25. Find the particular solution to the differential equation $\frac{du}{dt} = \tan u$ that passes through $(1, \frac{\pi}{2})$, given that $u = \sin^{-1}(e^{C+t})$ is a general solution.
26. Find the particular solution to the differential equation $\frac{dy}{dt} = e^{(t+y)}$ that passes through $(1, 0)$, given that $y = -\ln(C - e^t)$ is a general solution.
27. Find the particular solution to the differential equation $y'(1 - x^2) = 1 + y$ that passes through $(0, -2)$, given that $y = C\frac{\sqrt{x+1}}{\sqrt{1-x}} - 1$ is a general solution.

For the following problems, find the general solution to the differential equation.

28. $y' = 3x + e^x$
29. $y' = \ln x + \tan x$
30. $y' = \sin x e^{\cos x}$
31. $y' = 4^x$
32. $y' = \sin^{-1}(2x)$
33. $y' = 2t\sqrt{t^2 + 16}$
34. $x' = \coth t + \ln t + 3t^2$
35. $x' = t\sqrt{4 + t}$
36. $y' = y$
37. $y' = \frac{y}{x}$

Solve the following initial-value problems starting from $y(0) = 1$ and $y(0) = -1$. Draw both solutions on the same graph.

38. $\frac{dy}{dt} = 2t$
39. $\frac{dy}{dt} = -t$
40. $\frac{dy}{dt} = 2y$

41. $\frac{dy}{dt} = -y$

42. $\frac{dy}{dt} = 2$

Solve the following initial-value problems starting from $y_0 = 10$. At what time does y increase to 100 or drop to 1?

43. $\frac{dy}{dt} = 4t$

44. $\frac{dy}{dt} = 4y$

45. $\frac{dy}{dt} = -2y$

46. $\frac{dy}{dt} = e^{4t}$

47. $\frac{dy}{dt} = e^{-4t}$

Recall that a family of solutions includes solutions to a differential equation that differ by a constant. For the following problems, use your calculator to graph a family of solutions to the given differential equation. Use initial conditions from $y(t = 0) = -10$ to $y(t = 0) = 10$ increasing by 2. Is there some critical point where the behavior of the solution begins to change?

48. [T] $y' = y(x)$

49. [T] $xy' = y$

50. [T] $y' = t^3$

51. [T] $y' = x + y$ (Hint: $y = Ce^x - x - 1$ is the general solution)

52. [T] $y' = x \ln x + \sin x$

53. Find the general solution to describe the velocity of a ball of mass 1 lb that is thrown upward at a rate a ft/sec.

54. In the preceding problem, if the initial velocity of the ball thrown into the air is $a = 25$ ft/s, write the particular solution to the velocity of the ball. Solve to find the time when the ball hits the ground.

55. You throw two objects with differing masses m_1 and m_2 upward into the air with the same initial velocity a ft/s. What is the difference in their velocity after 1 second?

56. [T] You throw a ball of mass 1 kilogram upward with a velocity of $a = 25$ m/s on Mars, where the acceleration of gravity is $g = -3.711$ m/s². Use your calculator to approximate how much longer the ball is in the air on Mars than on Earth, where $g = -9.8$ m/s².

57. [T] For the previous problem, use your calculator to approximate how much higher the ball went on Mars, where $g = -3.711$ m/s².

58. [T] A car on the freeway accelerates according to $a = 15 \cos(\pi t)$, where t is measured in hours. Set up and solve the differential equation to determine the velocity of the car if it has an initial speed of 50 mph. After 40 minutes of driving, what is the driver's velocity?

59. [T] For the car in the preceding problem, find the expression for the distance the car has traveled in time t , assuming an initial distance of 0. How long does it take the car to travel 100 miles? Round your answer to hours and minutes.

60. [T] For the previous problem, find the total distance traveled in the first hour.
61. Substitute $y = Be^{3t}$ into $y' - y = 8e^{3t}$ to find a particular solution.
62. Substitute $y = a \cos(2t) + b \sin(2t)$ into $y' + y = 4 \sin(2t)$ to find a particular solution.
63. Substitute $y = a + bt + ct^2$ into $y' + y = 1 + t^2$ to find a particular solution.
64. Substitute $y = ae^t \cos t + be^t \sin t$ into $y' = 2e^t \cos t$ to find a particular solution.
65. Solve $y' = e^{kt}$ with the initial condition $y(0) = 0$ and solve $y' = 1$ with the same initial condition. As k approaches 0, what do you notice?